WORKSHEET · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 10: The Game Loop and Collisions

Module 4 | 45 minutes | Game loop and collision detection | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. game loop ___ A. Values that describe the game right now (lives, score).
b. collision ___ B. Reuse an object by moving it back to the start.
c. recycle ___ C. Two objects overlapping.
d. state ___ D. The loop that updates the game every frame.

2. Plan your code

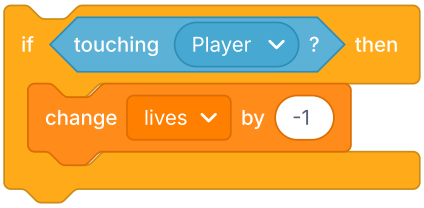
Before you build it: Endless rocks: the Rock falls forever (change y by -6). When y position < -170, set y to 170 and set x to pick random -40 to 40, so it falls again. Write or draw the blocks you will use.

3. Plan your code

Before you build it: Collisions: the Player is ready. Make lives start at 3. In the Rock’s loop, if it is touching the Player, change lives by -1 and set y to 170. Write or draw the blocks you will use.

4. Debug the collision

One hit takes all 3 lives at once. Which fix is best? Circle one.



a. A. Move the rock back to the top after the hit
b. B. Change lives by -0.3
c. C. Add a wait 5 seconds

5. Explain your thinking

Describe the game loop for the rock dodger in steps. Which parts happen every frame, and which only when a condition is true?

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can build a game loop that moves and recycles objects.
☐ I can detect collisions and update lives.
☐ I can stop one hit counting many times.